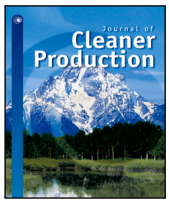
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How to transport carbon dioxide with minimal environmental impacts today and tomorrow? A prospective life cycle assessment

Julian Nöhl, Johannes Burger, Pauline Oeuvray, Viola Becattini, Jan Seiler, David Y. Shu, Marco Mazzotti, André Bardow *

Institute of Energy & Process Engineering, ETH Zurich, Tannenstrasse 3, Zurich, 8092, Switzerland

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ABSTRACT

Carbon capture, utilization, and storage is crucial for climate change mitigation. As carbon dioxide (CO_2) source and sink locations typically do not coincide, efficient CO_2 transport solutions are essential. Since installing transport infrastructure requires substantial investments, researchers and decision-makers need to know the long-term evolution of the resulting environmental impacts, and the implications for a gigatonne-scale deployment of CO_2 transport. Here, we quantify the environmental impacts of CO_2 transport modes in Europe up to 2100 via prospective life cycle assessment. Our results confirm that dense phase pipelines result in the lowest environmental impacts for both on- and offshore CO_2 transport today and in the future, despite substantial improvements in truck, train, barge, and ship transport towards 2100. Even for a gigatonne-scale CO_2 pipeline transport network in Europe, all 16 studied environmental impact categories are expected to be less than 0.075% of Europe's share of the safe operating space based on population. After pipelines, trains and electric trucks impact the climate the least. However, batch-wise onshore transport can exacerbate freshwater ecotoxicity, particulate matter formation, and metal/mineral depletion. Switching from batch-wise to dense phase pipeline transport can pay back the required carbon investment within less than three years. Our analysis, thus, identifies environmentally preferable CO_2 transport modes that are robust for future deployment towards gigatonne-scale carbon capture, utilization, and storage supply chains.

1. Introduction

The Paris Agreement aims to limit the average global temperature rise to below 1.5 °C, compared to preindustrial levels, by achieving a balance between anthropogenic greenhouse gas emissions and removal (UNFCCC, 2016). To stay below the 1.5 °C threshold, anthropogenic CO_2 emissions must reach net-zero by approximately 2050. Achieving net-zero emissions will require substantial emission reductions, circular carbon approaches, and carbon dioxide removal to offset residual emissions (IPCC, 2022).

Carbon capture can contribute to achieving climate goals in several ways: For hard-to-abate sectors such as the cement, steel, and chemical industries, permanent storage of CO_2 can be a key emission reduction solution (Paltsev et al., 2021). For carbon dioxide removal, scalable technologies include the permanent storage of CO_2 captured directly from the air or from biogenic sources (Bui et al., 2018; Minx et al., 2018). Circular carbon approaches include the utilization of the CO_2 to produce valuable products, such as chemicals, fuels, or plastics, contributing to the de-fossilization of the chemical industry (Kätelhön et al., 2019). Additionally, CO_2 can be stored in long-lived products, such as building materials produced through mineral carbonation of alkaline wastes (IPCC, 2005; Di Maria et al., 2020). As a result, the International Energy Agency (IEA) estimates carbon capture deployment in Europe to grow to the gigatonne-scale by the mid-century (IEA, 2020).

To enable the expansion of carbon capture, utilization, and storage technologies, efficient CO_2 transport is necessary because CO_2 point source locations are distributed and do not coincide with sink locations (IEA, 2020). For instance, the first large-scale European storage hubs are being developed in the North Sea or Iceland, including the Northern Lights storage hub in Norway (Equinor, 2019a), the Porthos project in the Netherlands (Schiferli, 2023), and the Carbfix Coda terminal in Iceland (Carbfix, 2024). As a result, designs of pioneering carbon capture, transport, and storage (CCTS) chains transporting CO_2 to those storage hubs include transport distances of up to several thousand kilometers (Oeuvray et al., 2024). Transport distances can be reduced by increasing the availability of storage locations and

* Corresponding author.
E-mail address: abardow@ethz.ch (A. Bardow).

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placing flexible direct air capture or utilization plants close to sinks or sources, respectively. However, assessments from the IEA reveal that approximately one-third of European emitters are over 100 km away from potential storage locations (IEA, 2020). Thus, substantial amounts of CO_2 will still need to be transported over long distances (IEA, 2020).

Pipelines are usually considered the preferred transport mode due to their good performance in terms of costs, economies of scale, and reliability (Oeuvray et al., 2024). Onshore, pipelines are the least-cost option for large-scale CO_2 transport, while offshore, fossil fuel-based ships can be competitive for longer distances (IPCC, 2005; Oeuvray et al., 2024). A comprehensive cost assessment of current CO_2 transport modes was conducted by Oeuvray et al. (2024).

However, the choice of CO_2 transport modes is not driven by cost alone but requires a multi-criteria assessment to determine the most suitable option for a specific application (Oeuvray et al., 2024). In particular, environmental sustainability should be a key consideration to avoid burden shifting from climate change mitigation to other impacts on the environment. The environmental impacts of CO_2 transport can be assessed via the life cycle assessment (LCA) methodology, quantifying climate change impacts and potential burden-shifting to other environmental impact categories. Most LCA studies assessing CCTS value chains assume CO_2 transport by pipeline (Corsten et al., 2013; Cuéllar-Franca and Azapagic, 2015; Cruz et al., 2021; Wang et al., 2022), suggesting low environmental impacts of transport. However, pipelines are usually not available today and the construction of a pipeline network will require years or decades, substantial capital investments, and multilateral agreements for cross-border transport (Skovholt, 1993; ZEP, 2019).

Until pipelines are available, CCTS value chains must rely on batch-wise transport modes, such as trucks and rail for land transport, and barges and ships for inland and maritime waterways (Burger et al., 2024; Oeuvray et al., 2024). Furthermore, batch-wise transport may even be required when large-scale pipeline infrastructure is available to connect small-scale CO_2 emitters, emitters at remote locations, or small-scale storage sites to the pipeline network.

Burger et al. (2024) assessed the environmental impacts of existing batch-wise transport modes in pioneering CCTS chains that could be deployed today, revealing substantial environmental impacts of batch-wise transport. For the CCTS chain connecting a CO_2 emitter in Switzerland to a geological storage site in the North Sea, transport dominates more than half of the impact categories, with climate change impact reaching up to 152 $kg_{CO_2\text{-eq.}}$ per $t_{CO_2\text{-transported}}$. Consequently, the transport step was identified as a critical target for improving CO_2 avoidance efficiency, particularly through the adoption of pipeline transport (Burger et al., 2024).

In summary, environmental impacts of CO_2 transport have been assessed for both currently available batch-wise transport and unavailable large-scale transport infrastructure using static or historical life cycle inventory data. However, according to IEA projections, most carbon capture plants will be constructed and operated between 2035 and 2070 (IEA, 2020), a period in which the transition of Europe's economy towards climate neutrality is ongoing, redesigning the energy and industrial system (European Commission and Directorate-General for Climate Action, 2019). Thus, the environmental impacts of CO_2 transport need to be estimated for this period to inform design decisions and avoid stranded assets due to unnecessary transport mode switches.

In this study, we quantify the environmental impacts of CO_2 transport modes throughout the scale-up phase of carbon capture, utilization, and storage. By incorporating scenarios from integrated assessment models (IAM) into the life-cycle inventory databases via the prospective LCA framework *premise* (Sacchi et al., 2022), we assess how choices for CO_2 transport are affected by the broader net-zero transition of the electricity system, cement and steel production, and the fuel markets.

We assess how the climate change impact of CO_2 transport modes can be expected to evolve for Europe up to 2100, investigating multiple IAM scenarios to address the uncertainty inherent in these projections (Section 3.1). We extend our analysis of climate change impact of transport modes by (1) discussing the influence of CO_2 sink conditions and transport distances on the optimal pipeline configuration (Section 3.2) and (2) assessing the benefits of switching to lower-impact transport modes based on the carbon payback period (Section 3.3). Furthermore, we extend our investigation of climate change impacts by 15 additional environmental impact categories and a discussion of the implications of a potential gigatonne-scale deployment of CO_2 transport in Europe (Section 3.4). To assess whether gigatonne-scale deployment of CO_2 transport is environmentally sustainable in absolute terms, we use life cycle impact assessment (LCIA)-based planetary boundaries.

2. Methods

In this study, we apply LCA methodology to assess the environmental impacts of CO_2 transport (ISO, 2020a,b). The following describes the goal and scope of the study (Section 2.1), the foreground models for CO_2 transport (Section 2.2) and conditioning (Section 2.3), the prospective background data (Section 2.4), and defines the carbon payback period (Section 2.5) and LCIA-based planetary boundaries (Section 2.6). The carbon payback period is used to assess the climate benefits by switching from existing transport infrastructure to low-impact transport modes and the LCIA-based planetary boundaries to quantify the environmental impacts of a gigatonne-scale transport deployment in absolute terms.

2.1. Goal and scope definition

The goal of the prospective LCA is to assess the environmental impacts of CO_2 transport modes in Europe up to 2100 to inform decision-makers. CO_2 transport is essential to connect sources and sinks in CCTS chains that consist of the steps:

- Carbon capture: separating CO_2 from point sources or the air,
- Transport conditioning: processing CO_2 to the state required for transport,
- Transport: moving CO_2 from source to sink,
- Storage conditioning: processing CO_2 to the state required for storage,
- Storage: permanently storing CO_2, e.g., in geological formations.

Our prospective LCA study distinguishes two system boundaries: System boundary T, including only the transport step, and system boundary T+C, including, additionally, the conditioning steps for transport and storage (Fig. 1). Furthermore, we distinguish between batch-wise and pipeline transport. Batch-wise transport includes both the transport of the full CO_2 vessel to the sink and the transport of the empty vessel back to the source, while pipeline transport only includes the one-way transport of CO_2.

System boundary T neglects conditioning to focus solely on the transport step. Hence, environmental impacts for system boundary T are independent of the CO_2 source and sink. System boundary T serves the function to transport conditioned CO_2 from the capture plant to the sink, with a functional unit of 1 ton of conditioned CO_2 transported over 1 km (= 1 tkm). CO_2 enters and leaves the system boundary T at defined conditions, either in the gaseous, liquid, or dense phase, dependent on the transport mode. The transport steps differ for batch-wise and pipeline transport (Fig. 1).

The chosen transport mode and the CO_2 source and sink specifications change the conditioning step. Specifically for the comparison of dense and gas pipelines, the conditioning step can impact the prospective LCA results. Hence, in **System boundary T+C**, we extend the system boundary to include the conditioning step in a comparison of dense and gas phase pipelines. Pure CO_2 is assumed to enter the system boundary at 1 bar and 30 °C and leave at the required sink conditions.

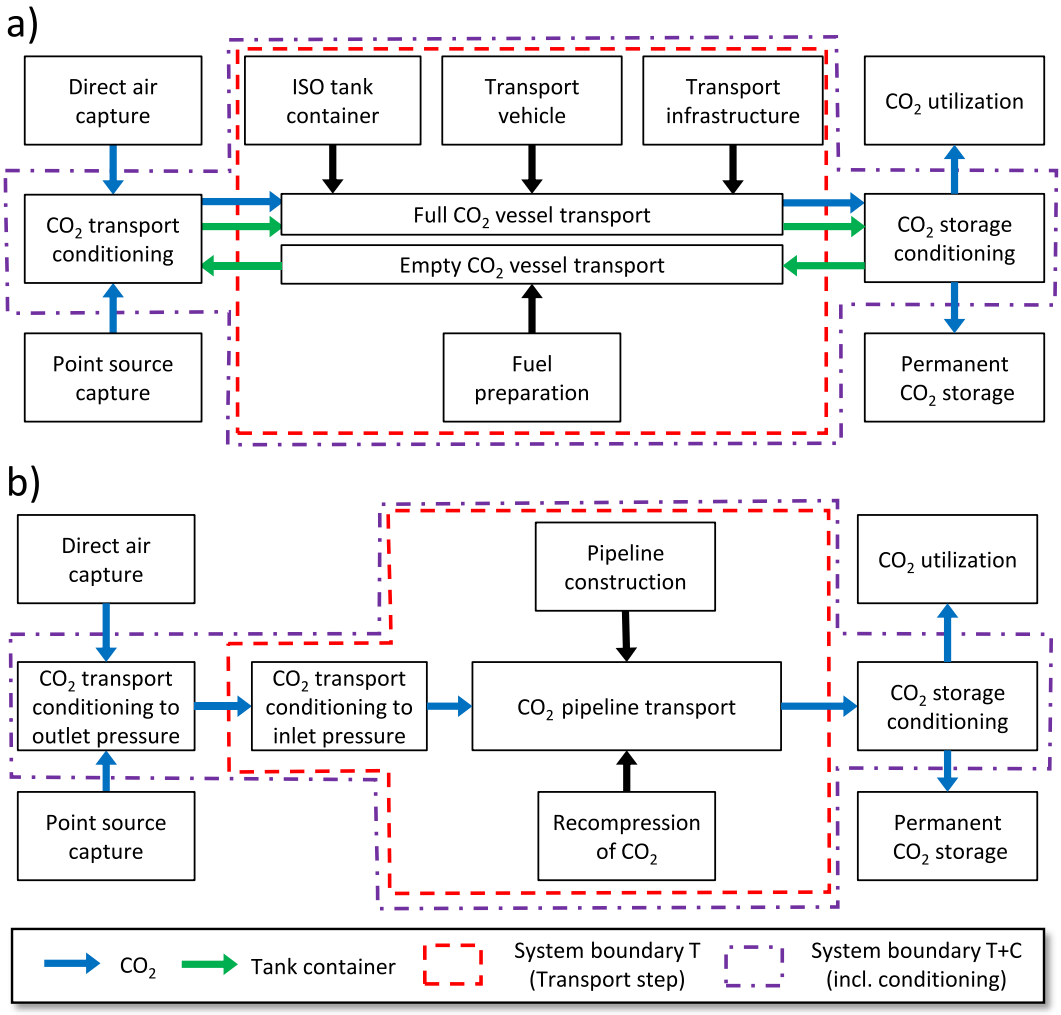


Fig. 1. System boundary for (a) batch-wise CO_2 transport and (b) CO_2 pipeline transport. In (b), the transport conditioning step is divided into two parts based on the pressure of the pipeline inlet and outlet. The pressure increase from pipeline outlet to inlet pressure is required to overcome the pressure loss in the pipeline and is thus allocated to system boundary T.

The function corresponding to system boundary T+C is the transport of unconditioned CO_2 from the capture plant to the sink, including conditioning for storage. The functional unit is 1 ton of unconditioned CO_2 transported over 1 km.

2.2. Foreground system: transport models

The considered transport modes for CO_2 include batch-wise transport via truck, train, barge for inland waterways, and ship for maritime waterways, as well as continuous transport via pipelines. Before transport, CO_2 is compressed or liquefied to decrease transport volume and cost. For batch-wise transport, CO_2 is liquefied at 16 bar and temperatures slightly below the boiling point (Oeuvray et al., 2024), whereas for pipelines, the CO_2 can be in a gaseous state at 15–30 bar or dense phase at 80–350 bar, both at ambient temperatures (Knoope et al., 2014).

Batch-wise transport assumes insulated ISO (International Organisation for Standardisation) tank containers, which can be transferred between carriers. The inventory data for these tank containers is based on manufacturer data (Meeberg Group, 2022). Each tank container weighs 9.2 t and has a capacity of 20.7 t of liquid CO_2 at 16 bar. Here, we conservatively assumed that gaseous CO_2 at the maximal allowable pressure remains in the tank container after offloading, resulting in 4% of the CO_2 remaining in the tank containers for the return trip.

Inventories for truck transport are generated using the tool *carculator_truck* (Sacchi et al., 2021), which produces inventory data for various power train technologies, truck size classes, and driving cycles. We investigate diesel, battery electric, and fuel cell electric trucks. The weight of the trailers used for loading the ISO tank containers is adjusted according to manufacturer data (Krone Commercial Vehicles SE, 2021), and the truckload mass is adjusted to the weight of the full and empty tank containers. Transporting one tank container requires a truck with a gross weight of approximately 44 t, including the total weight of the truck, driver, and freight. Battery electric trucks exceed a gross weight of 44 t in 2020 and 2025 but fall below this weight after 2025 due to the battery efficiency improvements considered by *carculator_truck*. *carculator_truck* distinguishes between three driving cycles defining the target speed for every second of driving. The driving cycles and the gross weight are used to calculate the fuel consumption of the vehicle and the emission of pollutants. Here, the *regional delivery* driving cycle with a range autonomy of 400 km is assumed. For comparison, we also assess the *urban delivery* and the *long-haul* driving cycles with a range autonomy of 150 km and 800 km, respectively.

Inventory data for transport by electric trains, diesel barges, and heavy fuel oil container ships are taken from the ecoinvent cutoff database version 3.8 (Wernet et al., 2016). As alternatives to heavy fuel oil ships, we include both methanol and ammonia ships. Methanol is assumed to be produced with CO_2 from a direct air capture plant

with heat pumps as the heat source and H_2 from water electrolysis, based on the inventory data from Hank et al. (2019), which is already integrated into the prospective LCA framework *premise* (Sacchi et al., 2022). Ammonia is assumed to be produced using the Haber-Bosch process with H_2 from water electrolysis and N_2 from air separation, based on inventory data for ammonia production from Mayer et al. (2023). Inventory data for H_2 from water electrolysis and N_2 from air separation is taken from *premise* to align assumptions between methanol and ammonia production. For both methanol and ammonia ships, we assume the same tank-to-wheel efficiency as for heavy fuel oil, with required fuel demand calculated based on lower heating values (Zincir and Arslanoglu, 2024). Inventory data on ship construction is based on heavy fuel oil container ships from ecoinvent. Ship exhaust emissions for methanol are taken from the International Maritime Organization (IMO, 2016) and for ammonia from Schwarzkopf et al. (2023).

For on- and offshore pipeline transport, we consider transport in both gaseous and dense phases. We assume a pipeline length of 200 kilometers and a mass-flow rate of 0.3 and 10 Mt CO_2 per year. To derive inventories, we determine pipeline configurations using the optimization algorithm of Oeuvray et al. (2024) and follow the methodology of ecoinvent for natural gas pipelines. The pipeline optimization algorithm determines the pipeline configuration with the lowest levelized cost for transporting CO_2 for a specified CO_2 mass-flow rate and pipeline length. The pipeline configuration includes the inner and outer pipeline diameter, the inlet pressure, the number of recompression stations, and the electricity demand for recompression. The optimized inlet pressure of the pipeline can vary between 15 and 30 bar for gas pipelines and 80 and 350 bar for dense pipelines. Thus, the inlet pressure can substantially exceed the outlet pressure of 15 bar for gas pipelines and 80 bar for dense pipelines. While system boundary T+C considers the full transport conditioning, system boundary T only considers the pressurization that is required to overcome the pressure loss in the pipeline (Fig. 1, bottom). Thus, we slightly modify the optimization algorithm to allocate the CO_2 pressurization between the conditioning and transport step (Supplementary Material A).

The steel demand for pipeline construction is derived from the pipe geometry. Similarly, the demand for coating materials for corrosion protection is estimated based on the pipe geometry, assuming the same coating material and thickness as in ecoinvent's natural gas pipelines. Pipelines are assumed to be operated at ambient temperatures of 15 °C for onshore and 4 °C for offshore pipelines and, thus, do not require any insulation material (Oeuvray et al., 2024). Sand demand for the burial of the onshore pipelines is calculated based on trench geometry recommendations (Bahadori, 2017; Koen et al., 2012). Additional inventory data is taken from ecoinvent's natural gas pipelines and linearly scaled with the pipeline cross-section area. Details on the pipeline model are provided in Supplementary Material A.

For comparison, we include an additional 8 Mt/y supercritical pipeline from *premise* based on the model from Wildbolz (2007). The CO_2 in supercritical state has a lower density but also a lower viscosity compared to CO_2 in dense phase.

The Supplementary Information B details the inventory data for all CO_2 transport modes.

2.3. Foreground system: conditioning model

System boundary T+C considers both the CO_2 conditioning to the state required for transport and the conditioning from the transport outlet conditions to the state required for geological storage. For transport conditioning, we assume that pure CO_2 enters at 1 bar and 30 °C, neglecting any additional purification steps. Thus, the capture step is assumed to provide CO_2 at the purity required for transport and storage.

Currently, geological storage formations for CO_2 primarily consist of saline aquifers, depleted oil and gas fields, and mineralization in basalts (Rycroft et al., 2024), resulting in differing required sink conditions. We distinguish between two sink conditions:

(1) Gaseous CO_2 at 25 bar: These conditions represent the injection condition for the Coda-terminal of Carbfix in Iceland, focusing on mineralization in basalt (Matter et al., 2011; Rycroft et al., 2024), and the receiving conditions of the Porthos storage hub in the Netherlands, which involves a depleted gas field (Schiferli, 2023; Rycroft et al., 2024).
(2) CO_2 in a dense phase at 70 bar and temperatures above 1 °C: These conditions represent the pipeline inlet conditions for the saline aquifer of the Phase 1 Northern Lights storage hub in Norway (Equinor, 2019b,a).

The energy requirements for transport conditioning to liquefied CO_2 at 16 bar are taken from Deng et al. (2019), who identified a least-cost 2-step compression and refrigeration process. Energy requirements for other conditioning processes are derived from process simulations using AspenPlus v11.1 (AspenTech, 2018). In particular, we consider the energy demand for compressors, pumps, cooling, refrigeration, and heating. We follow the same assumptions regarding property methods, the efficiency of components, and compression ratios as in Deng et al. (2019) and validated our process model by reproducing the energy requirements of the liquefaction process from Deng et al. (2019) with an accuracy of $\pm$ 4%. We assume surface water to be available for cooling and assess the environmental impacts of heating, cooling, and refrigeration using the heat pump, water pump, and refrigeration machine models from ecoinvent. The corresponding flowsheets of the conditioning processes can be found in Supplementary Material A. More details regarding the inventories and climate change impact of the conditioning processes are included in the Supplementary Material B.

2.4. Prospective background system

LCA typically applies static background databases based on present or past inventory data, e.g., ecoinvent (Wernet et al., 2016). To estimate future impacts of CO_2 transport, we apply the open-source Python framework *premise* (Sacchi et al., 2022) to the ecoinvent version 3.8 database. The *premise* framework modifies the ecoinvent database to reflect changes in the energy and industrial system in line with scenarios from IAMs. IAMs combine energy system models, economic system models, and climate science models to assess feasible pathways towards achieving climate goals (Hare et al., 2018; IPCC, 2022). This study uses three scenarios from the IAM REMIND v2.1: *SSP2-NDC*, *SSP2-Base*, and *SSP2-PkBudg500* (Baumstark et al., 2021). All three scenarios follow the shared socioeconomic pathway SSP2: 'Middle of the Road' narrative, in which social, economic, and technological trends follow historical patterns. The *SSP2-NDC* scenario follows the nationally determined contributions in the next decades and extrapolates policies with similar stringency. The *SSP2-Base* scenario is a baseline scenario without any implementation of climate policies. The *SSP2-PkBudg500* scenario includes a stringent global carbon budget of 500 Gt of CO_2 up to 2100.

We assess the environmental impacts of CO_2 transport in 5-year time steps between 2020 and 2050 and in 10-year time steps up to 2100 for each scenario. REMIND v2.1 aggregates countries into 12 global regions (Baumstark et al., 2021). In this study, we apply the prospective inventories from the European region (EUR), which includes all EU countries and the United Kingdom of Great Britain and Northern Ireland. Thus, the resulting environmental impacts represent the average European transport process and could deviate for individual countries.

2.5. Carbon payback period

Switching from existing to lower-impact transport modes reduces operational emissions but requires building new transport infrastructure. The carbon payback period Δt^{cpb} is the time after which the cumulative climate change impact of the new transport mode is lower

than that of continued use of the existing transport infrastructure (Eq. (1)).

$$\int_{t^s}^{t^s+\Delta t^{cpb}} I^{op,old}(t)\,dt \overset{!}{=} I^{inv,new}(t^s) + \int_{t^s}^{t^s+\Delta t^{cpb}} I^{op,new}(t)\,dt \quad (1)$$

Here, t^s is the switching year, $I^{inv,new}(t^s)$ the climate change impact from the construction of the new infrastructure in the switching year t^s, and $I^{op,old}(t)$ and $I^{op,new}(t)$ the use phase climate change impact in year t for the existing and the new transport infrastructure, respectively.

To calculate the carbon payback period, we discretize Eq. (1) by assuming constant climate change impacts within each year. The climate change impacts for each year were calculated using linear interpolation.

2.6. LCIA-based planetary boundaries

To assess whether the deployment of CO_2 transport on a gigatonne-scale is environmentally sustainable in absolute terms, we compare the environmental impacts of CO_2 transport to Europe's share of the safe operating space. By staying within the safe operating space, the probability of remaining in a Holocene-like state and, thereby, avoiding potentially unacceptable Earth System changes is maximized (Ryberg et al., 2020). Here, we quantify the safe operating space by LCIA-based planetary boundaries developed for 15 impact categories of the Environmental Footprint LCIA methods (Sala et al., 2020). The LCIA-based PBs represent the upper limit of the safe operating space. For the Environmental Footprint method *land use: soil quality index*, no LCIA-based planetary boundary is available. Thus, we use the current global environmental impact for this impact category (Andreasi Bassi et al., 2023).

The global safe operating space must be distributed among regions or industries by different allocation principles, which can strongly influence the conclusions (Ryberg et al., 2020). Following the study from Ryberg et al. (2018) and Sala et al. (2020), we allocated a 7.9% share of the global LCIA-based planetary boundaries to Europe based on its share of global population excluding Russia using the equality allocation method (United Nations, 2023).

We assume a yearly transport demand of 70 Gtkm for a future gigatonne-scale CO_2 transport network in Europe based on the projected increase of CO_2 capture rate to 0.7 Gt CO_2 per year in 2070 and an average proximity of CO_2 point sources to potential storage sites of 100 km following scenarios from the IEA (IEA, 2020).

3. Results

Here, we present the prospective environmental impact of CO_2 transport. Section 3.1 discusses the specific climate change impact for CO_2 transport per tkm in system boundary T, excluding conditioning. The extended system boundary T+C considering the conditioning step is discussed in Section 3.2. The carbon payback period to switch from existing transport infrastructure to lower climate change impact transport options is presented in Section 3.3. In Section 3.4, we extend our assessment beyond the climate change impact by 15 impact categories. Additionally, we quantify the environmental impact of gigatonne-scale CO_2 transport in Europe.

3.1. Climate change impact of CO_2 transport

In 2020, the climate impacts can already be very low with 4 gCO_2-eq. per tkm for pipelines (Fig. 2). The impacts are low primarily due to two factors: First, there are few greenhouse gas emissions from the operation, with low electricity requirements for repressurizing of CO_2 at recompression stations, and second, the large transport volume of pipelines over the lifetime results in low specific impacts from construction.

However, climate impacts can also be more than 40 times higher when employing trucks, primarily for two reasons: First, trucks have the highest energy consumption per tkm, due to a high driving mass-to-cargo ratio, as every ISO tank container requires a truck with a gross weight of approximately 44 t. The need for an additional return trip of empty ISO tank containers further increases fuel consumption. Second, the construction emissions per tkm are substantial due to the different components needed (ISO tank container, road infrastructure, truck) and the lower lifetime of the trucks compared to pipelines with multi-decade lifetime. Consequently, the embedded construction emissions per tkm transported by truck are as high as 38 gCO_2-eq. for diesel trucks, 43 gCO_2-eq. for hydrogen trucks, and 85 gCO_2-eq. for electric trucks in 2020.

The prospective assessment shows that, by 2100, all climate change impacts are expected to decline by 72–93%. The largest absolute climate impact reduction is observed for batch-wise transport, with an average reduction of 127 gCO_2-eq./tkm for truck technologies. However, despite the steep reductions, the impacts remain higher than those of large pipelines, whose emissions are already low today and for which the relative fall of impacts is even higher with an 86% reduction (vs. 81% for truck transport). For transport modes other than pipelines, the climate change impact ranking changes until 2030 and stabilizes afterward. This changing ranking should be considered in the long-term planning of CCTS networks whenever pipelines are unavailable.

The development of climate change impacts follows three characteristic trends depending on the fuel source:

(1) Electricity- or e-fuels-based transport modes undergo a substantial reduction in climate change impact between 2020 and 2035 regardless of the IAM scenario (Supplementary Material Figure A.10 and A.11), due to the decarbonization of electricity with minor differences in decarbonization speed between scenarios. The climate change impact reduction rate is closely linked to the electricity demand of each transport mode, with higher relative reductions observed for modes with greater reliance on electricity. The climate change impacts of electric trains and trucks fall well below the impact of diesel barges and diesel trucks, which are competitive today only due to the high-greenhouse gas electricity grid mix. Similarly, the climate change impacts of methanol- and ammonia-fueled ships are initially higher than those of heavy fuel oil ships due to the greenhouse gas intensity of the electricity for electrolysis. However, impacts of both methanol and ammonia-fueled ships quickly fall well below the impacts of conventional heavy fuel oil ships for a decarbonized energy system. After 2035, climate change impacts continue to fall slowly for all electricity or e-fuels-based transport modes due to the falling greenhouse gas footprint of steel, cement, and some carbon-based fuels.

(2) Transport modes using diesel-like fuels, such as trucks and barges, have high initial climate change impacts that steadily decline to comparable levels with trains and electric trucks in the scenarios *SSP2-NDC* and *SSP2-PkBudg500* towards 2100. The decline of climate change impact is due to an assumed increasing share of biofuels and e-fuels. Without the decarbonization of the diesel market mix in the pessimistic scenario *SSP2-Base*, the climate change impact of diesel-based transport remains high (Supplementary Material Figure A.11).

(3) As *premise* does not include a prospective heavy fuel oil market mix, the climate change impact of heavy fuel oil ships remains constant regardless of the IAM scenario.

In summary, on- and offshore CO_2 pipelines impact the climate the least regardless of the IAM scenario and year. As CO_2 pipelines exist in the US and are comparable to widespread natural gas pipelines in Europe, we consider the technology to be mature and confidence in inventories to be high. The low climate impacts of large-scale pipelines align with their economic advantages, as literature identifies pipelines as the most cost-effective CO_2 transport option (Oeuvray et al., 2024).

For CCTS value chains without access to pipeline infrastructure, low climate change impact alternatives are electric trains and ships fueled by e-fuels. Today, these transport technologies have comparable climate change impacts to diesel barges and heavy fuel oil ships but are expected to have much lower climate change impacts soon. However,

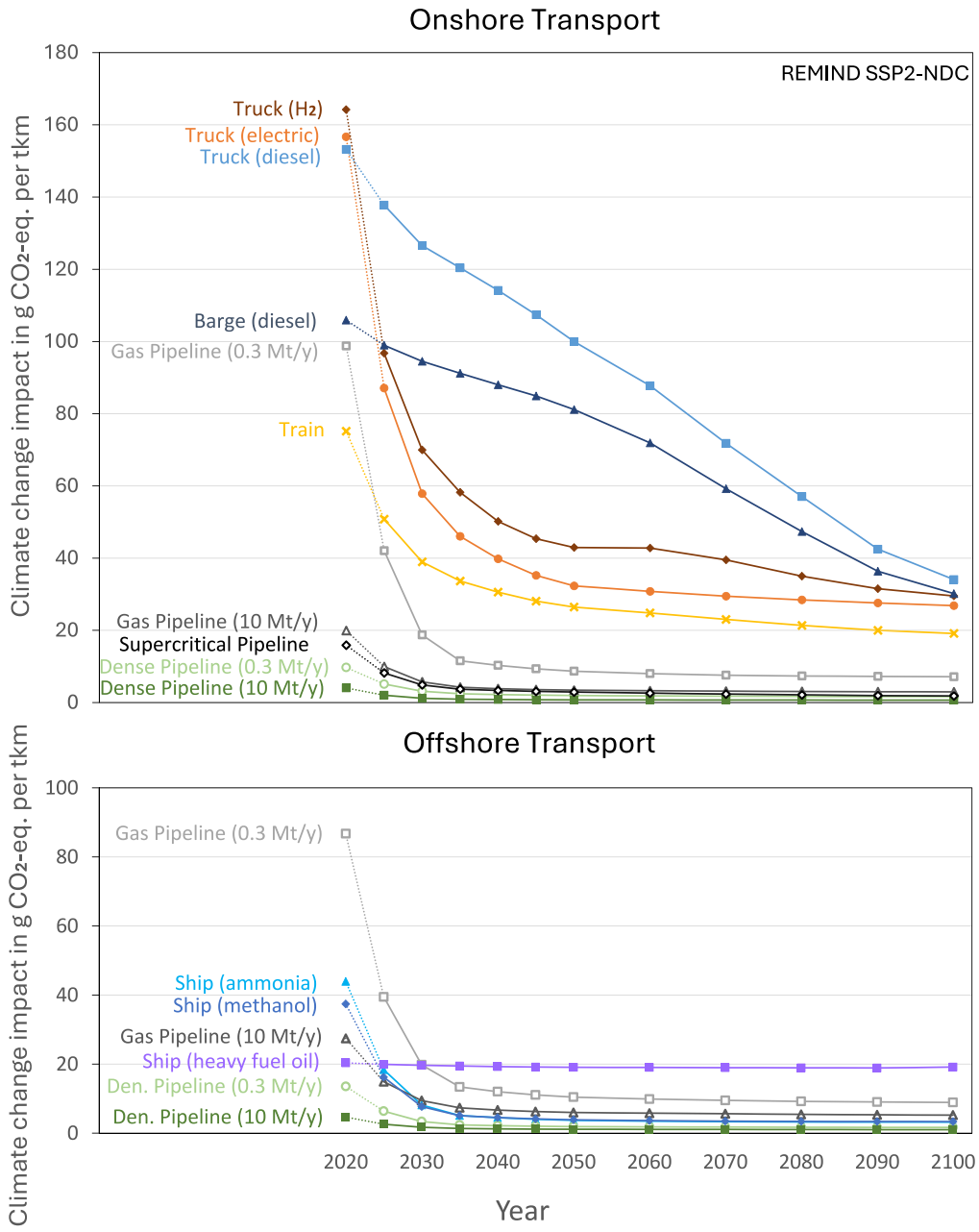


Fig. 2. Climate change impact of the onshore (top) and offshore (bottom) transport for 1 ton CO_2 over 1 km between 2020 and 2100 for the Remind *SSP2-NDC* scenario following the nationally determined contributions and system boundary T, which excludes conditioning. All life cycle phases take place in the same year. The dotted lines illustrate that the starting year of Remind is 2020.

inventory data for the production and use phases of e-fuels are subject to higher uncertainty.

Truck transport impacts the climate more than train transport but might be necessary when connections to rail- or waterways are unavailable. Electric and H_2 trucks are expected to have a lower climate change impact than diesel trucks, regardless of the assumed range autonomy (Supplementary Material Figure A.9). However, the environmental impacts of battery production are subject to considerable uncertainty. In particular, variations in the supply chains for battery production, specifically the country of production and the sources of raw materials, may result in substantial differences in impacts.

Moreover, a trade-off exists between environmental and economic performance: Recent cost estimates indicate that for transport distances above 500 km, CO_2 transport via barges may be more cost-effective than trains, while for short distances, trucks tend to be the most economical batch-wise transport mode (Oeuvray et al., 2024). Balancing these environmental and economic considerations is therefore essential.

3.2. The climate change impact of conditioning for pipeline transport

The energy required for CO_2 conditioning varies substantially across transport modes. Specifically for gas and dense phase pipelines, the conditioning step can change the climate change impact ranking of pipeline transport.

For storage sites requiring gaseous CO_2 at 25 bar, the conditioning steps impacts the climate less for gaseous pipelines than for dense phase pipelines. As transport distances increase, the relative contribution of the conditioning step decreases. Consequently, dense phase pipelines

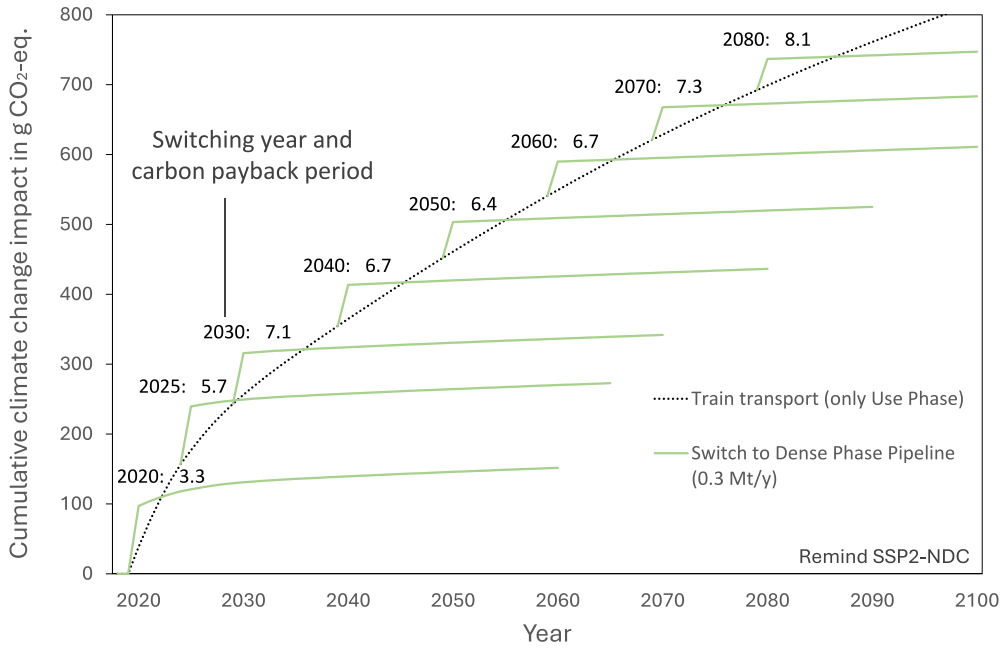


Fig. 3. Cumulative climate change impact for transporting 1 tkm of CO_2 by existing train transport infrastructure (dotted black line) and for switching to dense phase pipeline (continuous green lines) for the *SSP2-NDC* scenario and system boundary T. The carbon payback period varies depending on the switching year as indicated in the figure. A pipeline lifetime of 40 years is assumed.

have a lower climate change impact than gaseous pipelines for longer transport distances (Supplementary Material Figure A.13). In 2040, this threshold distance is 125 km for a capacity of 0.3 Mt CO_2 per year and 340 km for a capacity of 10 Mt per year.

For CO_2 storage sites requiring CO_2 in dense phase at 70 bar pressure or higher, dense phase pipelines impact the climate least regardless of the transport distance (Supplementary Material Figure A.12). The higher energy demand for compression to dense phase at the pipeline inlet is compensated by the reduced energy demand for subsequent compression required for geological storage.

Liquefaction for batch-wise transport modes is energy-intensive and results in a higher climate change impact compared to conditioning for pipeline transport (Supplementary Material B). Thus, conditioning further strengthens the preference for pipelines over batch-wise transport from a climate perspective.

3.3. Carbon payback period of switching to low-impact transport modes

Pipelines and e-fuel-based transport modes outperform conventional transport modes from an environmental perspective but might be unavailable for pioneering CCTS chains. Once available, switching from conventional to lower-impact transport modes reduces operational emissions but requires building new transport infrastructure. To assess the climate benefit of switching transport modes, we calculate the *carbon payback period* Δt^{cpb}, following the concept of the economic payback period (Paltrinieri and Khan, 2016).

Fig. 3 illustrates the carbon payback period for switching from an existing train transport infrastructure to a dense phase pipeline with a capacity of 0.3 Mt per year for different switching years. The initial steep increase in the green line reflects the emissions from the pipeline's construction, followed by a substantially reduced slope during use phase. The carbon payback period ranges from 3.3 years if the switch is made today to 8.1 years if made in 2080, indicating that construction emissions are offset within a few years. The increasing carbon payback period towards 2080 is due to the decreasing use phase emissions and the increasing relative importance of construction emissions over time.

The carbon payback period for switching to low-impact transport modes depends on the combination of existing and new transport infrastructure (Fig. 4). For the investigated 66 combinations, the year 2040 was selected as a reference point, as it represents a plausible time when established pioneering infrastructure could switch to low-impact transport modes. For 49 of 66 investigated combinations, switching to low-impact transport modes in 2040 reduces cumulative climate change impact, with carbon payback periods ranging from less than 6 months to 38 years. For the remaining 17 transport mode combinations, switching to low-impact transport modes increases the cumulative climate change impact and should thus be avoided.

Switching from batch-wise transport modes to pipelines is beneficial in most cases, with carbon payback periods below seven years or even below three years for dense phase pipelines (Group A in Fig. 4). However, there are exceptions: (1) switching from electric trucks to pipelines other then dense phase pipelines with large capacities (10 Mt/y); (2) switching from trains or heavy fuel oil ships to gas pipelines; and (3) switching from ships using electricity-based fuels to dense phase pipelines leads to long carbon payback periods of 14 to 31 years (Group B).

For diesel or heavy fuel oil-based transport modes, switching to pipelines results in climate benefits in less than 3 years. Switching from diesel or heavy fuel oil-based to electricity-based batch-wise transport modes can also reduce cumulative climate change impacts in less than 13 years (Group C).

As the climate change impact is similar across pipeline types (Fig. 2), switching between pipelines rarely makes sense and can increase cumulative climate change impact (Group D). The carbon payback periods fall only below 11 years when the pipeline size can be substantially increased and when switching to a more efficient dense phase pipeline.

3.4. Environmental impacts beyond climate change

While sustainability discussions around CCTS often focus on climate change mitigation, a holistic assessment is essential to account for potential burden-shifting to other environmental impact categories. We assessed and ranked on- and offshore CO_2 transport modes across 15 environmental impact categories beyond climate change, using the Environmental Footprint 3.0 assessment methods (Fazio et al., 2018; European Commission, 2019).

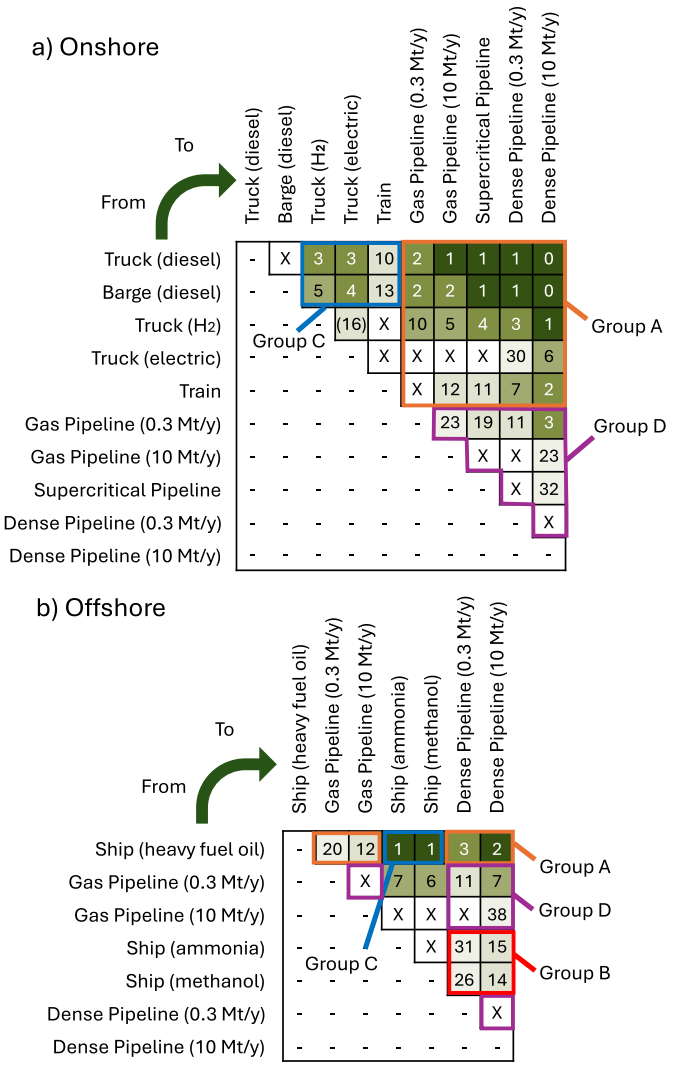


Fig. 4. Carbon payback period (in years) for switching from an existing transport infrastructure (rows) to a new transport mode (columns) in (a) onshore and (b) offshore transport in 2040, considering the Remind *SSP2-NDC* scenario and system boundary T. Cases where switching to lower-impact transport modes increases cumulative climate change impact due to small use phase emissions of the existing transport infrastructure are indicated by *X*. The number in parentheses represents a case where the carbon payback period is larger than the lifetime of the transport modes. For the analysis presented in the main text, the results are grouped based on the characteristics of the transport mode switch.

Dense phase pipelines lead to the lowest environmental impacts across all 16 environmental impact categories for onshore and 14 out of 16 categories for offshore transport (Fig. 5). Thus, dense phase pipelines are found to be the most environmentally sustainable CO_2 transport mode for both onshore and offshore transport.

The second-best transport mode after dense phase pipelines differs for onshore and offshore transport: For onshore transport, gas phase pipelines outperform all batch-wise transport modes. For offshore transport, however, gas pipelines only perform better than ships in some impact categories.

Batch-wise transport modes show substantial variability across impact categories, necessitating trade-off analyses to guide decision-making. For example, electric trucks have a low climate change impact when powered by low-carbon electricity. However, electric trucks may cause higher impacts than H_2 trucks in 11 out of 16 categories and higher impacts than diesel trucks in 4 out of 16 categories, largely due to the resource- and emission-intensive nature of battery production. Similarly, ships powered by heavy fuel oil impact the climate more than ships using alternative fuels but perform better in half of the other environmental impact categories. For ships powered by alternative fuels, many of these impacts are driven by the high electricity demand of H_2 production via water electrolysis.

These examples underscore the importance of evaluating trade-offs across multiple impact categories, particularly for batch-wise transport modes.

To evaluate the absolute sustainability of CO_2 transport at a gigatonne scale, we compare the environmental impacts of a future European CO_2 transport demand of 70 Gtkm (cf. Section 2.6) with Europe's share of the safe operating space within planetary boundaries.

For context, road freight in the EU-27 totaled 1800 Gtkm in 2020, contributing 0.6% to 1.9% to gross value added (GVA) in major European economies (Zanne et al., 2023). Following the GVA-based allocation approach by Ryberg et al. (2018) and assuming the same ratio between Gtkm and share of GVA, a 0.023% to 0.074% share of safe operating space could be allocated to the CO_2 transport sector.

Environmental impacts of gigatonne-scale CO_2 transport span several orders of magnitude, ranging from 0.000003% to 4.8% of Europe's share of the safe operating space for onshore transport modes and 0.000005% to 0.46% for offshore transport modes (Fig. 5). This variability highlights the potential to prioritize the impactful categories in sustainability assessments.

Impact categories with substantial contributions should be prioritized in decision-making. In our case, these categories are *freshwater ecotoxicity*, *metal/mineral depletion*, and *particulate matter formation*. Categories with negligible contributions, such as *ozone depletion* or *ionizing radiation*, appear less critical for selecting sustainable CO_2 transport modes.

Dense phase pipelines remain favorable, with gigatonne-scale impacts contributing less than 0.075% of Europe's share of the safe operating space across all impact categories. The highest contributions occur in *freshwater ecotoxicity*, *metal/mineral depletion*, and *particulate matter formation* and range from 0.012% to 0.075%, depending on the pipeline capacity. Other impact categories remain below 0.023% of Europe's share of the safe operating space.

In contrast, batch-wise onshore transport modes potentially lead to major environmental impacts in the *particulate matter formation*, *freshwater ecotoxicity*, and *metal/mineral depletion* category, ranging between 0.4% and 4.8% of Europe's share of the safe operating space. Thus, batch-wise onshore transport modes can strongly exceed the allocated share of the safe operating space following the GVA-based approach.

Transport via diesel barges and trucks may cause impacts of more than 0.075% of Europe's share of the safe operating space in up to 10 impact categories and electric trains, electric trucks, and H_2 trucks in up to 7 categories.

For offshore transport, the environmental impacts remain below 0.075% of Europe's share of the safe operating space for most impact categories and transport modes, suggesting that offshore transport modes are environmentally preferential.

Overall, the assessment of environmental impact categories beyond climate change further underpins the environmental superiority of pipelines over batch-wise transport modes, in particular for dense phase pipelines. A gigatonne-scale dense phase pipeline network can achieve absolute environmental sustainability by operating within its share of the safe operating space. Among batch-wise onshore transport modes, trains impact the environment the least but might still transgress the allocated share of the safe operating space in up to 7 impact categories. Learning not accounted for in the prospective LCA might improve the situation. Still, this analysis highlights the need to balance trade-offs among environmental impact categories.

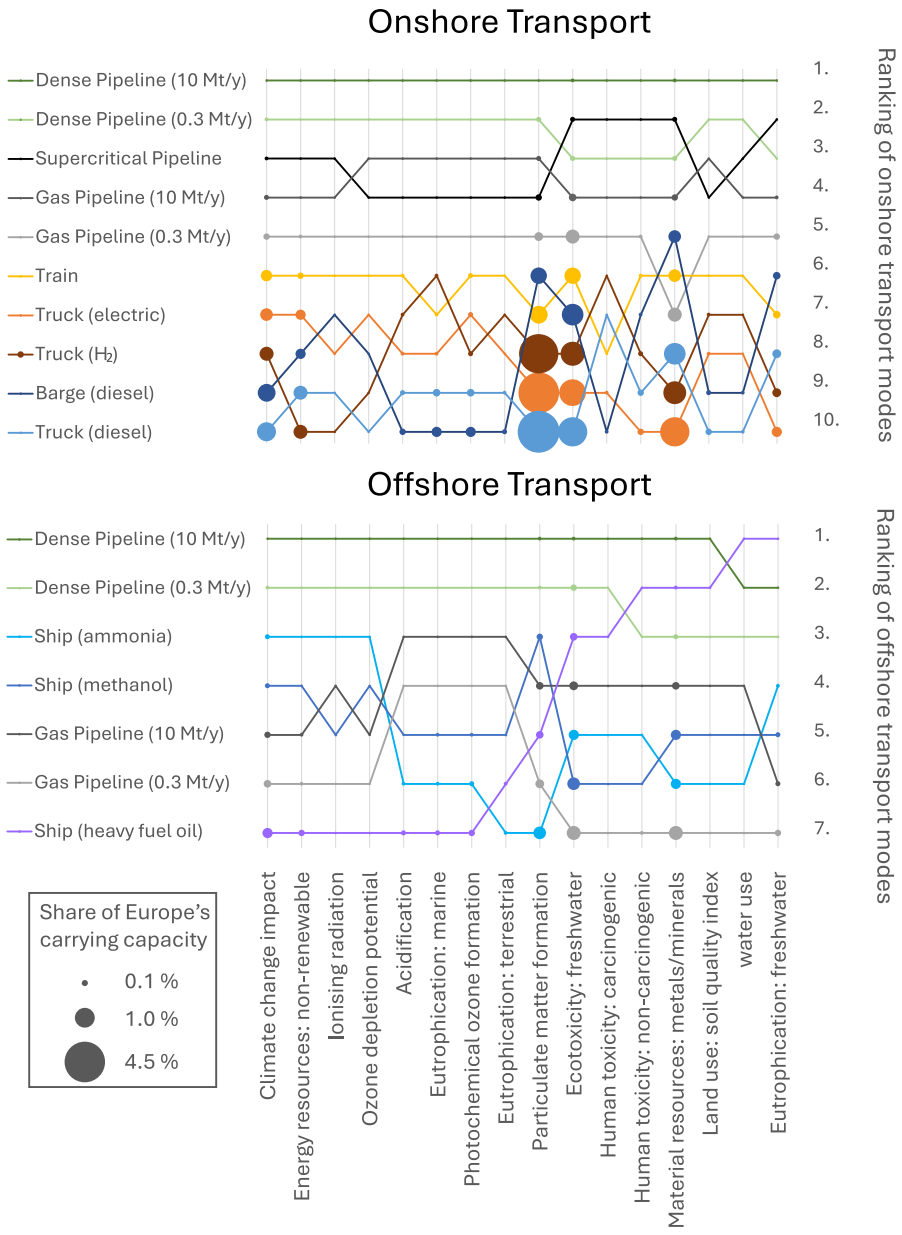


Fig. 5. Environmental impact ranking of onshore (top) and offshore (bottom) transport modes for the Remind *SSP2-NDC* scenario and system boundary T. The 1st place represents the transport method with the lowest environmental impact in the corresponding impact category. All phases of the life cycle take place in 2070. The circle area represents the impact of a gigatonne-scale deployment of transport relative to Europe's share of the safe operating space.

4. Conclusion

The conducted prospective life cycle assessment shows that CO_2 pipelines exhibit the lowest climate change impacts for on- and offshore transport across all scenarios and time steps. Today, climate change impacts of pipelines can already be very low with 4 gCO_2-eq. per tkm. However, transport impacts can also be more than 40 times higher when employing trucks. By 2100, climate change impacts of CO_2 transport modes are expected to decline by 72–93% compared to current levels. Despite these substantial reductions, the ranking of climate change impacts is stable across all prospective scenarios as early as 2030.

Transport via dense phase pipelines leads to lower climate change impact than gas phase pipelines. Only when storing gaseous CO_2 at 25 bar requiring a conditioning step and when transport distances are short, gas phase pipelines impact the climate less than dense phase pipelines.

When pipeline infrastructure is unavailable, low climate change impact alternatives include electric trains and e-fuel ships. While these transport modes have climate change impacts comparable to diesel barges and heavy fuel oil ships today, they are expected to substantially reduce their climate change impacts as low-carbon electricity becomes available. Truck transport might be necessary for CO_2 sources lacking connections to rail or waterways, but results in higher climate change impacts than trains. Electric trucks are expected to have lower climate change impacts than diesel trucks, regardless of the range autonomy.

The carbon payback period to assess the climate benefit of switching transport modes indicates that, in many cases, the construction emissions of the new transport infrastructure are offset within a few years, making the switch to low-impact transport beneficial. Switching from batch-wise transport modes to dense phase pipelines, in most cases, results in carbon payback periods of less than 3 years. However, switching between different pipeline types may even increase cumulative climate

change impacts unless the pipeline capacity is substantially increased and upgraded to a more efficient dense phase transport mode.

Beyond climate change, we assessed 15 environmental impact categories for a potential gigatonne scale deployment of CO_2 transport in Europe. Overall, dense phase pipelines have the lowest environmental impacts across all impact categories for both on- and offshore transport. Even a dense phase pipeline infrastructure at a gigatonne-scale is expected to have minor environmental impacts, remaining below 0.075% of Europe's share of the safe operating space. The minor environmental impacts of large-scale pipelines align well with their economic advantages identified in the literature.

In contrast, batch-wise onshore transport modes can lead to substantial environmental impacts in *freshwater ecotoxicity*, *particulate matter formation*, and *metal/mineral depletion*, with impacts ranging between 0.4% and 4.8% of Europe's share of the safe operating space. In contrast, environmental impacts for offshore transport remain below 0.075% for most impact categories and transport modes.

Our analysis shows that the broader transition outlined by IAM scenarios drastically reduces the climate change impact of CO_2 transport modes. However, our assessment also underscores the importance of careful technology selection to avoid burden-shifting among environmental impact categories. These insights provide guidance for optimizing the long-term planning and development of CCTS networks, ensuring environmental benefits across multiple dimensions.

CRediT authorship contribution statement

Julian Nöhl: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Johannes Burger:** Writing – review & editing, Validation, Methodology. **Pauline Oeuvray:** Writing – review & editing, Validation, Software, Methodology. **Viola Becattini:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Jan Seiler:** Writing – review & editing, Supervision, Conceptualization. **David Y. Shu:** Writing – review & editing, Supervision, Conceptualization. **Marco Mazzotti:** Writing – review & editing, Project administration. **André Bardow:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: AB served on review committees for research and development at ExxonMobil and TotalEnergies, oil and gas companies that are also active in carbon capture, transport, utilization and storage. AB has ownership interests in firms that render services to industry, some of which may work on carbon dioxide capture, transport, utilization and storage. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jclepro.2025.145659.

Data availability

The models used to create the life cycle assessment are included in the Supplementary Material. All required data can be licensed from Ecoinvent.

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